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Title: Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input,output, functions, call by reference in C++.

```
#include <iostream>
using namespace std;

void swap(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;
}

void read(int arr[], int &n) {
    cout << "Enter number of elements: ";
    cin >> n;
    cout << "Enter " << n << " numbers:\n";
    for(int i = 0; i < n; i++) {
        cin >> arr[i];
    }
}

void display(int arr[], int n) {
    for(int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}

void sort(int arr[], int n) {
    for(int i = 0; i < n - 1; i++) {
        for(int j = 0; j < n - i - 1; j++) {
```

```
            if(arr[j] > arr[j + 1]) {
                swap(arr[j], arr[j + 1]);
            }
        }
    }
}

int main() {
    int arr[100];
    int n;

    read(arr, n);

    cout << "Original array: ";
    display(arr, n);

    sort(arr, n);

    cout << "Sorted array: ";
    display(arr, n);

    return 0;
}
```